

Chapter 22: Normal Labor — Revision Notes

Williams Obstetrics 26e, pp. 1063–1101 of the PDF. High-yield numbers are in **bold**. This chapter has no tables. 10 of the 11 figures are included from the PDF (22-8 is omitted because it repeats 21-4 and 22-9).

Big picture: Labor = regular contractions → effacement and dilation → delivery of the newborn and placenta. It is physiological and normal for most women. Know **fetal orientation** (lie, presentation, attitude, position), the **7 cardinal movements**, the **Friedman vs Zhang labor curves**, and the practical management of the 1st and 2nd stages.

1. Fetal Orientation

Fetal lie

- = relation of the fetal long axis to the maternal long axis.
- **Longitudinal in >99%** of term labors. Transverse is less frequent.
- **Oblique lie** (axes cross at **45°**) is **unstable** → becomes longitudinal or transverse in labor.

Fetal presentation

- **Presenting part** = the fetal part in or nearest the birth canal.
- Longitudinal lie → **cephalic** or **breech**. Transverse lie → **shoulder**.

Cephalic type	Head/neck	Presenting part	Note
Vertex (occiput)	Sharply flexed , chin on thorax	Occipital (posterior) fontanel	Usual
Sinciput	Partly flexed	Anterior (large) fontanel	Transient → converts to occiput
Brow	Partly extended	Brow	Transient → converts to face
Face	Sharply extended , occiput touches back	Face	

- If sinciput or brow do not convert → **dystocia** (Ch 23).
- **Why the fetus presents head-first:** the **podalic pole** (breech + limbs) is bulkier and more mobile than the head, so it takes the roomier **fundus**.
- Breech falls with gestational age, but **2–3% of singletons are breech at delivery**. Types: **frank, complete, footling** (Ch 28).

Fetal attitude (habitus)

- Ovoid mass matching the uterine cavity: **convex back**, head sharply flexed, thighs flexed over the abdomen, knees bent; arms across the thorax; cord between the limbs.
- From occiput → face presentation, the spine goes from **convex (flexed) to concave (extended)**.

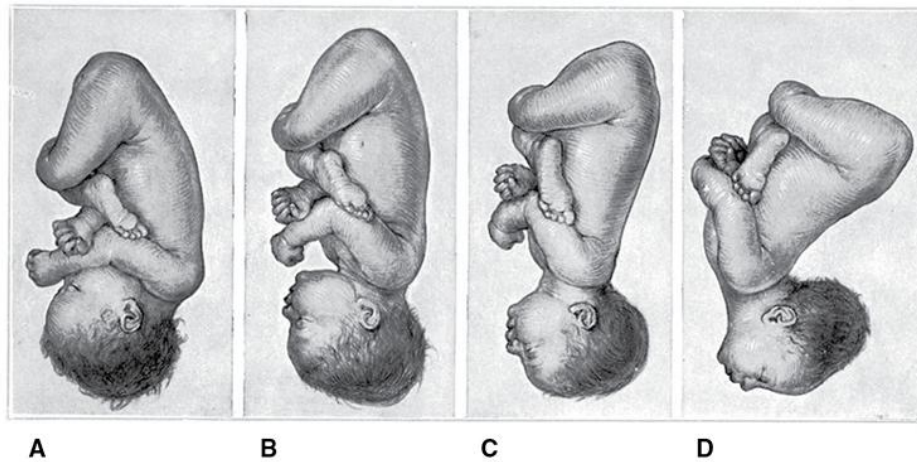


Figure 22-1. Attitude in (A) occiput, (B) sinciput, (C) brow and (D) face presentations: the head becomes progressively less flexed.

Fetal position

- = relation of a **defined landmark** of the presenting part to the maternal **right or left**, and to the **anterior, transverse or posterior** pelvis.

Presentation	Landmark	Positions
Occiput	Occiput	LOA, LOT, LOP, ROA, ROT, ROP, OA, OP
Face	Chin (mentum)	LM / RM $\pm$ A/T/P
Breech	Sacrum	LS / RS $\pm$ A/T/P
Shoulder	Acromion (scapula)	Right/left; dorsoanterior vs dorsoposterior

- Transverse lie **"back up" vs "back down"** matters for choosing the **cesarean incision**.

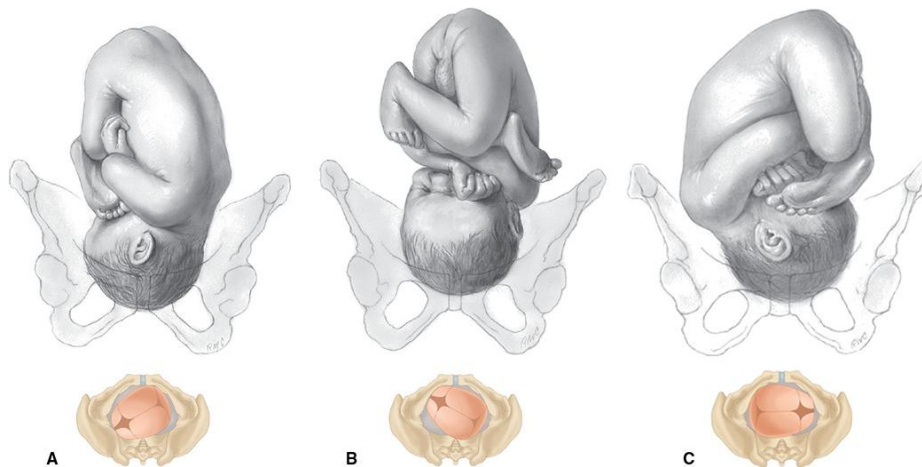


Figure 22-2. Occiput presentations: (A) LOA, (B) LOP, (C) ROT.

Diagnosis

Leopold maneuvers (1894) — mother supine. Hard in **obesity, polyhydramnios, anterior placenta**.

Maneuver	Assesses	Findings
1st	Fundus → lie and presentation	Breech = large, nodular ; head = hard, round
2nd	Sides → position (back vs limbs)	Back = hard, resistant (occiput on that side); limbs = small, irregular, mobile. OA → convex back felt ; OP → nodular limbs felt
3rd	Above the symphysis → presentation , engagement	Movable mass = not engaged
4th	Examiner faces the feet → descent	Fingertips slide along the inlet axis

- Accuracy for malpresentation: **sensitivity 88%, specificity 94%, PPV 74%, NPV 97%**.
- Fetal weight by palpation correlates **poorly**, especially in obesity.
- **Vaginal exam**: hard before labor (closed cervix). In labor, identify **sutures and fontanelles**; facial features → face; sacrum and perineum → breech; ribs, scapula, clavicle → transverse lie.
- **Sonography** confirms abnormal presentation/lie. OA: spine anterior, the spine–occiput angle widens as the head flexes. **OP: orbits and nasal bridge anterior. Sonography is more accurate than digital exam for head position in the 2nd stage.**

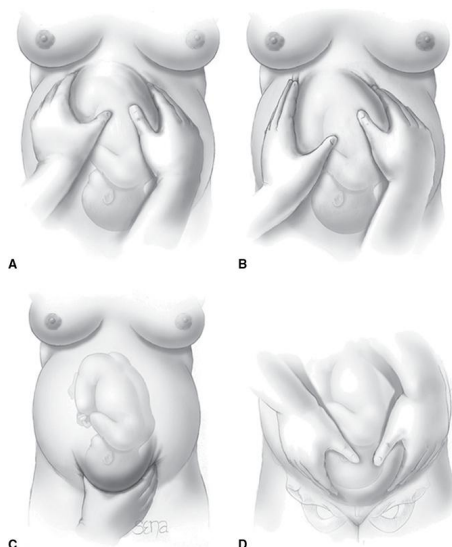


Figure 22-3. Leopold maneuvers A–D (fetus in LOA).

2. Mechanisms of Labor (Cardinal Movements)

Memory hook: "Every Diligent Fetus Is Eager Exiting Early" = Engagement, Descent, Flexion, Internal rotation, Extension, External rotation, Expulsion.

- The movements **overlap in time**. Contractions straighten the fetus (loss of dorsal convexity) → the ovoid becomes a **cylinder** with the smallest cross-section.

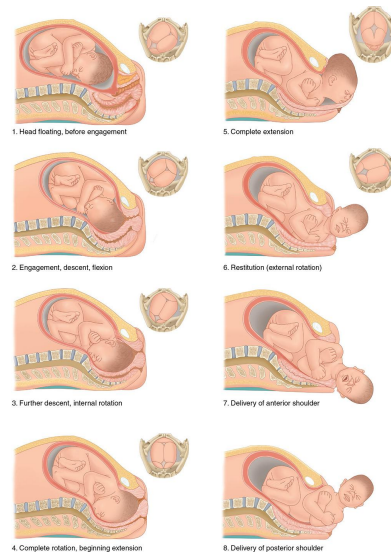


Figure 22-4. Cardinal movements from LOA: floating → engagement/descent/flexion → internal rotation → extension → restitution → anterior then posterior shoulder.

Engagement

- **Definition: the biparietal diameter passes through the pelvic inlet.**
- May occur in the last weeks of pregnancy or only after labor starts. **"Floating"** head at labor onset is common in multiparas and some nulliparas.
- In **5341 nulliparas**, no engagement before labor did **not** lower vaginal delivery rates (spontaneous or induced).
- Head usually enters with the **sagittal suture transverse**; **LOT** slightly more common than **ROT**.

Synclitism and asynclitism

	Sagittal suture lies near	Bone presenting
Synclitism	Midway between symphysis and promontory	—
Anterior asynclitism	Sacral promontory	Anterior parietal
Posterior asynclitism	Symphysis	Posterior parietal

- **Moderate asynclitism is normal**; successive shifting posterior → anterior asynclitism aids descent.
- **Severe** asynclitism → common cause of **CPD even with a normal pelvis**. Extreme → an **ear** is palpable.

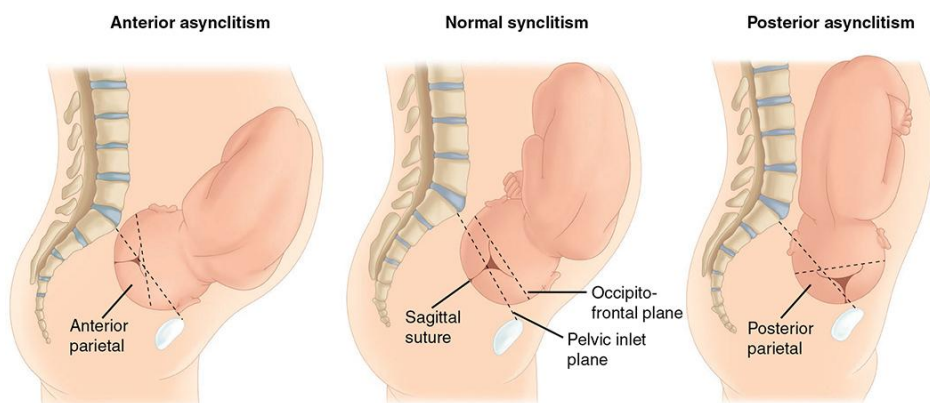


Figure 22-5. Anterior asynclitism (anterior parietal presents), synclitism, posterior asynclitism (posterior parietal presents).

Descent

- **First requisite for vaginal birth.**
- **Nullipara:** may engage before labor, then no further descent until the **2nd stage**. **Multipara:** descent usually begins **with** engagement.

- **Forces:** (1) fundal pressure on the breech during contractions, (2) maternal bearing-down, (3) extension/straightening of the fetal body.

Flexion

- When the head meets resistance (cervix, pelvic walls, floor) → chin to thorax.
- The shorter **suboccipitobregmatic** diameter replaces the longer **occipitofrontal** → essential for descent.

Internal rotation

- Occiput turns from transverse, **usually anteriorly** toward the symphysis (LOT → LOA; ROT → ROA). Less often posteriorly → OP.
- **Essential**, except for an unusually small fetus.
- **Calkins (>5000 women):**
 - **~2/3** complete rotation **by the time the head reaches the pelvic floor**
 - **~1/4** shortly after reaching the floor
 - **~5%** do not rotate
- If rotation waits for the floor: **1–2 contractions in multiparas; 3–5 in nulliparas.**

Extension

- Uterine force (posterior) + resistance of pelvic floor and symphysis (anterior) → resultant vector toward the **vulva** → extension.
- **Base of the occiput pivots under the inferior symphysis.** Born in order: **occiput, anterior fontanel, brow, nose, mouth, chin** over the perineum; then the chin drops over the anus.
- Without extension, the head would be forced through the posterior perineum.

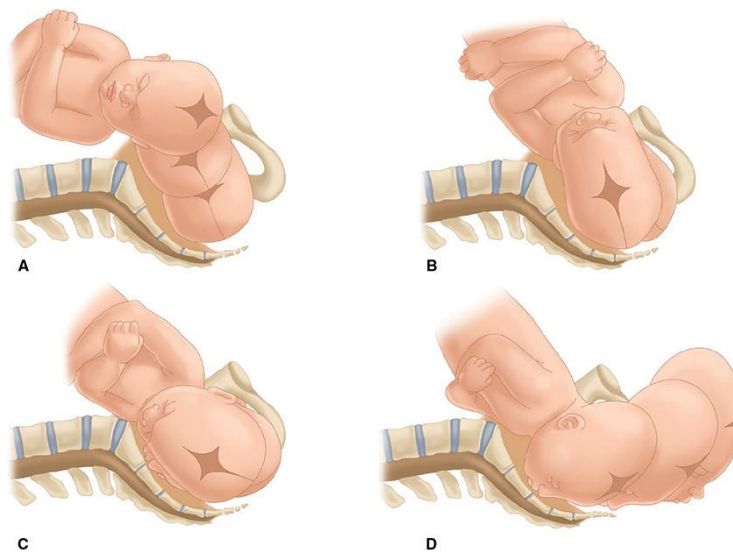


Figure 22-6. LOT mechanism: engagement with posterior asynclitism → anterior asynclitism → correction → internal rotation to OA and extension.

External rotation (restitution)

- Occiput turns back to its original side (toward the maternal **left or right ischial tuberosity**) → transverse.
- **Bisacromial diameter aligns with the AP diameter of the outlet** → one shoulder anterior (behind the symphysis), one posterior.

Expulsion

- Anterior shoulder under the symphysis → posterior shoulder distends the perineum → body follows.
- Anterior shoulder wedged behind the symphysis = **shoulder dystocia** (Ch 27).

Occiput posterior position

- Enters the pelvis OP in **~20%**; associated with a **narrow forepelvis**.
- Most rotate anteriorly at the pelvic floor (with good contractions, flexion and average size).
- **Incomplete or no rotation in 5–10%**, especially with a **large fetus**. Predisposed by poor contractions, faulty flexion, **epidural** (less pushing, relaxed pelvic floor).
- Outcomes: **transverse arrest** or **persistent OP** → dystocia, cesarean (management Ch 27; manual rotation Ch 29).

Fetal head shape changes

	Caput succedaneum	Molding
What	Scalp edema over the cervical os	Change in bony head shape from compression
When	Prolonged labor before full dilation; often low in the canal against a rigid outlet	Some before labor (Braxton Hicks); in labor
Features	Usually a few mm; may obscure sutures/fontanels. Over the most dependent part → reveals original position. Far from the midline with marked asynclitism	Parietal bones seldom overlap ("locking" at coronal and lambdoid sutures). Shortens suboccipitobregmatic diameter
Significance	—	May decide vaginal vs cesarean birth with a contracted pelvis or asynclitism. No proven link to neurological injury. Resolves within ~1 week

- Molding vs caput vs **cephalohematoma** → Ch 33.



Figure 22-7. (A) Caput far left of the midline (arrow) from marked asynclitism, after cesarean for arrest. (B) Caput and elongated molding after vaginal birth.

3. Normal Labor Characteristics

- **Strict definition:** contractions that cause **demonstrable effacement and dilation**. Onset is judged **retrospectively**; no consensus on definitions.
- Onset methods: (1) clock time when painful contractions become regular; (2) time of admission.
- **False labor** wanes spontaneously; true labor progresses to a regular effective pattern.
- **US admission criteria (intact membranes): dilation $\geq 3-4$ cm + frequent painful contractions.**

First stage — Friedman's sigmoid curve

Friedman division	Phase	Features
Preparatory	Latent	Little dilation, but big connective-tissue changes. Arrested by sedation and conduction analgesia
Dilatational	Active (maximum slope)	Fastest dilation. Unaffected by sedation
Pelvic	Deceleration + 2nd stage	Cardinal movements mainly here; onset seldom clear

- Active phase = **acceleration** → **maximum slope** → **deceleration**.

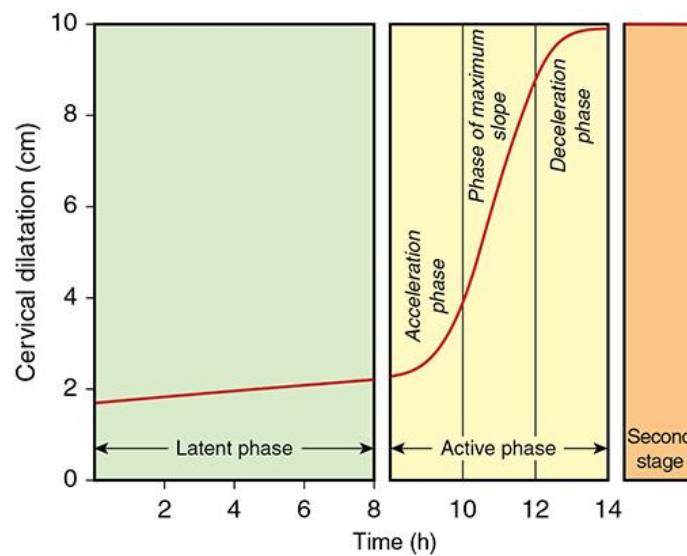


Figure 22-9. Average nulliparous dilatation curve: flat latent phase, then an active phase with acceleration, maximum slope and deceleration.

Latent phase

- Starts when the mother perceives **regular contractions**; for most, ends at **4 cm**.
- **⚠ ACOG/SMFM (2019): active labor now begins at 6 cm** (Ch 23).
- **Prolonged latent phase (Friedman): >20 h nullipara, >14 h multipara** (95th percentiles).
- Causes: **excess sedation or epidural, unfavorable cervix** (thick, uneffaced, undilated), **false labor**.
- Heavily sedated latent labor: **85%** enter active labor, **10%** stop (false labor), **5%** need oxytocin. → **Amniotomy discouraged** (10% false labor).
- Prolonged latent phase did not raise fetal/maternal morbidity or mortality (Friedman), but is linked to **longer active labor and more interventions**.

Active phase

	Friedman (1955, 1972)	Zhang (2010; 62,415 women)
Threshold	Slope changes at 3–5 cm ; 3–6 cm + contractions reliably = active labor	—
Nullipara duration	Mean 4.9 h (SD 3.4) ; statistical maximum 11.7 h	Median 4→5 cm 1.3 h , 5→6 cm 0.8 h , then ~0.5 h/cm
Dilation rate	Nullipara 1.2–6.8 cm/h (minimum 1.2); multipara minimum 1.5 cm/h	Normal may take >6 h for 4→5 cm and >3 h for 5→6 cm
Multiparas	Somewhat faster	Same as nulliparas 4–6 cm, then much faster

- **Descent** begins late in active dilation: from **7–8 cm in nulliparas**, fastest **after 8 cm**.
- Zhang's data underpin the **ACOG Obstetric Care Consensus (2019)** on cesarean for arrest (Ch 23).

Second stage

- Full dilation → delivery. **Median ~50 min nulliparas, ~20 min multiparas** (highly variable).
- High-parity women may need only **2–3 pushes**.
- **Longer** with a contracted pelvis, large fetus, conduction analgesia or sedation, and **high starting station** in nulliparas.
- **Higher BMI does not prolong** the 2nd stage.

Labor duration

Study	Nullipara	Multipara
Kilpatrick & Laros (1st + 2nd stage, no regional analgesia)	Mean ~9 h; 95th percentile 18.5 h	Mean 6 h; 95th percentile 13.5 h

- Onset = recalled regular painful contractions every **3–5 min** with cervical change.
- **Parkland (~25,000 spontaneous term labors): ~80% admitted at ≤5 cm**. Parity and admission dilation determine length. **Median admission → delivery 3.5 h; 95% delivered within 10.1 h** → normal active labor is relatively short.

4. Management of Normal Labor

- Guidelines for Perinatal Care (AAP/ACOG 2017) set personnel and facility standards. Out-of-hospital birth → Ch 27.

EMTALA

- Medicare-participating hospitals with emergency services must give an **appropriate screening exam** to any pregnant woman with contractions.
- **Labor** = from the **latent phase through delivery of the placenta**. A woman with contractions is **in true labor unless a physician certifies false labor** after reasonable observation.
- A woman in true labor is **"unstable" for transfer until newborn and placenta are delivered**. Transfer only at her request or if a physician certifies benefits outweigh risks.

Identification of labor

- Report early, especially if risk factors.
- **Advice rule: contractions 5 min apart for 1 h (≥12/h)** without ruptured membranes or bleeding → possible labor. Pates (768 women): active labor (≥4 cm) diagnosed **within 24 h in ¾** of those with ≥12/h.
- **Parkland triage (≥24 wk): intact membranes and <4 cm** → EFM for **up to 2 h**. Cervical change or persistent contractions → admit; otherwise home with false labor.
 - 3949 women discharged with false labor at 37–41^{6/7} wk: returned after a **mean 4.9 days; no ↑ adverse neonatal outcome or cesarean**.
- ACOG endorses **hospital-based obstetrical triage units**.
- **Latent labor** in low-risk women: **expectant care is reasonable** → ↓ oxytocin, antibiotics for fever and cesarean. Some need admission for pain. Agree a time to reassess.

Initial evaluation

- BP, temperature, pulse, RR; FHR by Doppler, sonography or fetoscope; review records.
- **Vaginal exam unless known placenta previa or vasa previa**. Index + middle fingers, avoid the anal region.

Ruptured membranes — why it matters

1. **Cord prolapse** if the presenting part is not fixed
2. **Labor likely soon** at or near term

3. **Infection** (intrauterine and neonatal) rises the longer delivery is delayed

Diagnosis of rupture

Test	Positive	Pitfalls
Sterile speculum	Fluid pooling in the posterior fornix or clear fluid from the os	—
pH	Vaginal 4.5–5.5 ; amniotic fluid >7.0	—
Nitrazine	pH >6.5	False +: blood, semen, bacterial vaginosis. False –: scant fluid
Ferning	Arborization on a dried slide (NaCl, protein, carbohydrate)	—
AFP in the vaginal vault	Present	—
Fluorescein (rare)	1–4 mL of 5% sodium fluorescein by amniocentesis	Invasive
Point-of-care: AmniSure (PAMG-1), Actim PROM (IGFBP-1), ROM Plus (IGFBP-1 + AFP)	Protein detected	False + and – not uncommon; FDA: combine with clinical findings

Cervical assessment

- **Dilation:** sweep the finger rim to rim; **full = 10 cm**.
- **Effacement:** canal length vs an unlabored cervix. Half the length = **50%**; as thin as the lower segment = **100%**.
- **Position:** posterior, mid, anterior (descent moves it anteriorly). **Consistency:** soft, intermediate, firm.
- **Station:** leading edge vs the **ischial spines** (midway between inlet and outlet = **midpelvis**). **ACOG (1989): fifths, each = 1 cm**, from **–5 → 0 → +5**.
 - **+5 = head visible at the introitus.**
 - At **0 station or below, the head is usually engaged** (biparietal plane through the inlet) — but **extensive molding/caput can give 0 station without engagement.**
- **Bishop score** = dilation, effacement, consistency, position, station → predicts induction success (Ch 26).

Laboratory

- Recheck **Hct/Hb** (Parkland point-of-care spun Hct in **3 min**; send to lab if **<30%**).
- **Blood type and antibody screen.**
- Some states (e.g., Texas) require **syphilis, hepatitis B and HIV** testing at admission even if done prenatally.
- Urine protein: Parkland tests **hypertensive women only**.

5. Management of First-Stage Labor

- Complete the general exam; plan monitoring. **Don't promise a labor duration.**

Fetal heart rate monitoring (AAP/ACOG 2017)

	First stage	Second stage
No risk factors (auscultate after a contraction, or review EFM)	Every 30 min	Every 15 min
At-risk pregnancy	Every 15 min	Every 5 min

Maternal monitoring

- **Temperature, pulse, BP at least every 4 h**; temperature **hourly** if membranes ruptured for many hours or borderline fever.
- Manual contraction assessment: at the acme of a firm contraction the uterus **cannot be indented**.
- **After membrane rupture:** if the head was not definitely engaged → **exam promptly to exclude cord prolapse**; check FHR now and through the next contraction (occult cord compression).

- Parkland: pelvic exams every **2–3 h** in active labor. Link between number of exams and infection is **conflicting**.

Maternal position

- Any comfortable position; often **lateral recumbency**. **Avoid supine** (aortocaval compression).
- Cochrane (Lawrence 2013): upright/ambulant → 1st stage **~1 h shorter**, less cesarean and epidural (variable-quality studies).
- **Bloom (1067 women, Parkland RCT): ambulation did not change labor length or analgesia need and was not harmful.** → Offer recumbency or supervised ambulation.

Pain management

- Patient-directed. Relaxation (weak evidence), ambulation, **water immersion** (Cochrane: ↓ regional analgesia, no ↑ adverse outcomes).
- Pharmacological: **epidural**, IV/IM **opioids or meperidine**, less often **nitrous oxide** (Ch 25).

Oral intake

- **No food or particulate liquids in active labor** — gastric emptying is **markedly prolonged** in labor and with analgesics → vomiting and **aspiration**.
- **Moderate clear liquids are fine** in uncomplicated labor: water, clear tea, black coffee, carbonated drinks, Popsicles, pulp-free juice.
- **Planned cesarean: stop liquids 2 h and solids 6–8 h before.**

IV fluids

- Not always needed. Useful for analgesia, fluid before regional block, and **postpartum oxytocin** (prevent/treat atony).
- Long labors: glucose + Na + water at **60–120 mL/h** prevents dehydration and acidosis.
- **Dextrose-containing fluids may shorten labor**; rates **125 or 250 mL/h** suitable. **Parkland: D5 saline at 150 mL/h.**

Amniotomy

- Pros: early detection of **meconium**; allows a scalp electrode or IUPC.
- **Does not shorten normal spontaneous labor** (meta-analysis of 15 studies).
- Risks: **infection** (if early) and **cord prolapse** → head must be well applied and not dislodged.
- **GBS unknown:** give intrapartum prophylaxis for **ROM >18 h or temperature >38.0 °C (100.4 °F)** (Ch 67). Also ↓ chorioamnionitis and endometritis.

Bladder

- Distention hinders descent → later **hypotonia and infection**. Palpate suprapubically; encourage voiding.
- **Catheterize** if distended and unable to void (**regional analgesia** is a common reason). **Continuous vs intermittent: similar UTI and retention rates.**

6. Management of Second-Stage Labor

- **Head position by vaginal exam:** fingers sweep from posterior forward over the head → cross and trace the **sagittal suture** → identify the fontanel at each end → then station. Use **sonography** if unclear.

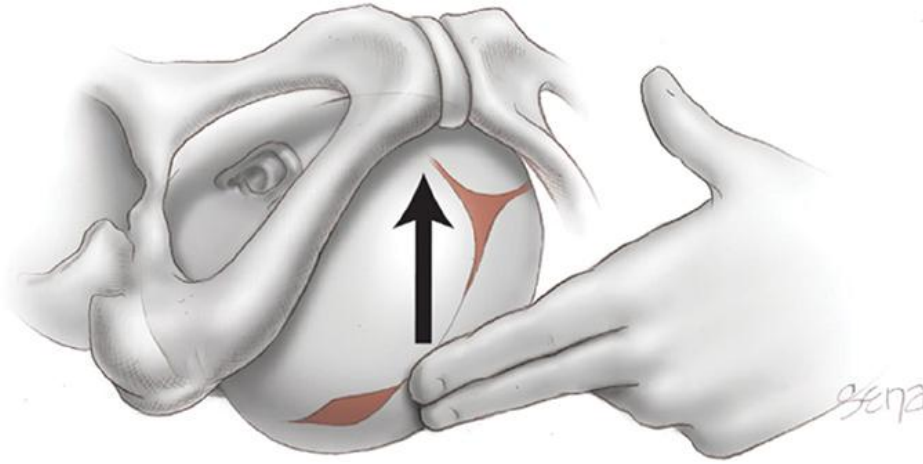


Figure 22-10. Locating the sagittal suture: sweep the fingers forward over the head toward the symphysis.

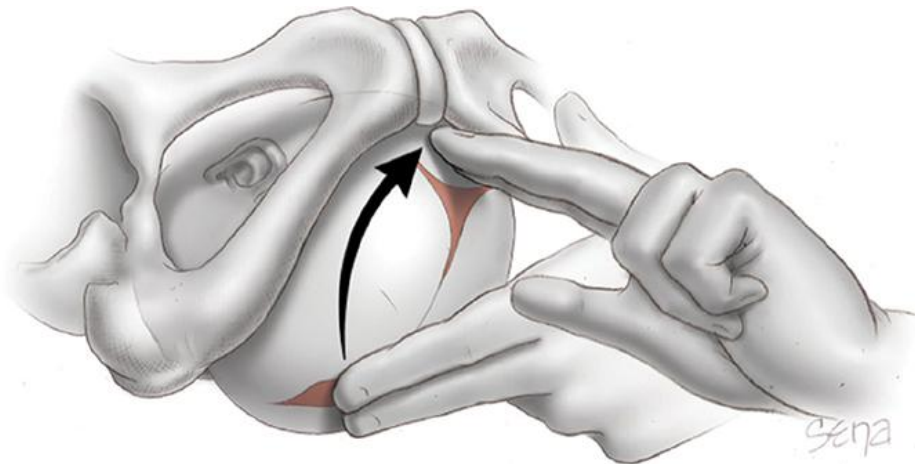


Figure 22-11. Differentiating the fontanelles by following the sagittal suture to each end.

- Full dilation → **urge to bear down / defecate**. Contractions last **~1 min**, at intervals **<90 s**.
- Median **50 min nulliparas, 20 min multiparas**.

Pushing

- Usually reflexive; may be **blunted by neuraxial analgesia**.
- **Delayed pushing (60 min)** proposed for passive descent. **RCT (Cahill 2018, nulliparas with neuraxial)**: delivery route **similar**; immediate pushing = **9 min longer pushing** but **less chorioamnionitis**. Meta-analysis agrees.
- Coach to push as if straining at stool **during each contraction only**; rest in between. FHR may slow during a push but **should recover before the next**.
- **Coached vs uncoached: same outcomes**.

Position

- **BUMPES trial (nulliparas with epidural)**: lateral recumbent (bed incline $\leq 30^\circ$) → **41% spontaneous birth vs 35% upright** (significant). OVD, anal sphincter tears and neonatal outcomes similar.
- Cochrane (with epidural): position has little effect on "operative birth".
- **Without epidural** (Cochrane, Gupta 2017): upright → slightly shorter, fewer episiotomies and OVDs, but **↑ blood loss >500 mL** and perhaps **↑ 2nd-degree tears**.
- **No position is mandated**; reposition for comfort and fetal trajectory.
- **Lithotomy**: avoid prolonged excessive hip **flexion, abduction or external rotation** → **femoral or lumbosacral nerve injury**.
- Transperineal ultrasound **angle of progression** may predict vaginal birth (early data).
- Perineal bulging and visible scalp → prepare for delivery (Ch 27).

7. Labor Management Protocols

Active management of labor (O'Driscoll, Dublin)

- Standardized protocol → cesarean rate **5%** (1970s–80s). Key components: **amniotomy + oxytocin**; constant **midwife** attendance.
- **Labor diagnosed**: painful contractions + **complete effacement, bloody show, or ruptured membranes** → committed to delivery **within 12 h**.
- Pelvic exam **hourly for 3 h**, then **every 2 h**.
- Dilation **<1 cm/h** → **amniotomy**; reassess in 2 h → **high-dose oxytocin** unless ≥ 1 cm/h (Ch 26).
- Membranes ruptured before admission → oxytocin for no progress at **1 h**.

Trial	Result
López-Zeno (705 nulliparas, Chicago)	Cesarean 10.5% vs 14.1% (active vs traditional), significant
Frigoletto (1934 nulliparas, Boston)	Labor somewhat shorter; no change in cesarean rate
Cochrane (Wei 2013)	Modest reduction in cesarean

Parkland protocol

- **Admit if active labor (≥ 4 cm + regular contractions) or confirmed ROM.**
- Pelvic exam **~every 2 h**.
- No dilation within **~2 h** of admission → **amniotomy** → reassess in 2 h.
- No progress → **IUPC** to assess contractions.
- **Hypotonic contractions + no dilation after 2–3 more hours** → **high-dose oxytocin**.
- **Goal: 200–250 Montevideo units for 4 h before diagnosing dystocia.**
- **Accepted progress on oxytocin: 1–2 cm/h**; may take **≥ 8 h** before cesarean for dystocia.
- Evaluated in **>20,000** uncomplicated pregnancies: did **not jeopardize the fetus-newborn**.

Quick-Recall: Normal Labor at a Glance

Topic	Key facts
Lie	Longitudinal >99% ; oblique (45°) is unstable
Presentation	Vertex usual; breech 2–3% at delivery; sinciput/brow transient
Landmarks	Occiput (vertex), mentum (face), sacrum (breech), acromion (shoulder)
Engagement	BPD through the inlet ; usually 0 station; sagittal suture transverse; LOT > ROT
Asynclitism	Anterior = suture toward promontory (anterior parietal presents); posterior = suture toward symphysis
Flexion	Suboccipitobregmatic replaces occipitofrontal
Internal rotation	$\frac{2}{3}$ by pelvic floor, $\frac{1}{4}$ just after, 5% none
OP	20% enter OP; 5–10% fail to rotate → transverse arrest / persistent OP
Caput vs molding	Caput = scalp edema; molding = bone shape; parietal bones seldom overlap
Latent phase	Ends at 4 cm (Friedman); active labor from 6 cm (ACOG 2019); prolonged >20 h / >14 h
Active phase	Friedman nullipara min 1.2 cm/h , multipara 1.5 cm/h ; Zhang: 4→5 cm may take >6 h
Second stage	Median 50 / 20 min
Nitrazine	pH >6.5 = ROM; false + blood, semen, BV
FHR checks	Low risk 30 / 15 min ; at risk 15 / 5 min (1st / 2nd stage)
GBS unknown	Prophylaxis for ROM >18 h or >38.0 °C
Parkland dystocia	200–250 MVU for 4 h before diagnosing dystocia

Key Numbers to Memorize

- Longitudinal lie **>99%**; breech **2–3%** of singletons at delivery; OP at entry **~20%**, no rotation **5–10%**
- Leopold: sens **88%**, spec **94%**, PPV **74%**, NPV **97%**
- Rotation at the floor: **1–2 contractions (multipara), 3–5 (nullipara)**
- Latent phase ends **4 cm**; active labor **6 cm** (ACOG/SMFM 2019); prolonged latent **>20 h nullipara, >14 h multipara**
- Sedated latent labor: **85% active / 10% false / 5% oxytocin**
- Friedman active phase nullipara **4.9 h (max 11.7 h), 1.2 cm/h** min; multipara **1.5 cm/h**
- Descent starts **7–8 cm**, fastest after **8 cm**
- 2nd stage median **50 min / 20 min**; total labor **9 h (18.5 h)** nullipara, **6 h (13.5 h)** multipara
- Parkland: median admission → delivery **3.5 h; 95% within 10.1 h**
- **≥12 contractions/h** → labor; triage EFM **up to 2 h** if <4 cm
- Vaginal pH **4.5–5.5**; amniotic fluid **>7.0**; nitrazine **>6.5**
- Station **–5 to +5 cm**; +5 = crowning
- Planned cesarean: **liquids 2 h, solids 6–8 h** NPO
- IV fluid **60–120 mL/h** (125–250 suitable; Parkland D5-saline **150 mL/h**)
- GBS: **>18 h ROM** or **>38.0 °C**
- BUMPES: recumbent **41%** vs upright **35%** spontaneous birth
- Active management: cesarean **5%** (Dublin); **10.5% vs 14.1%** (López-Zeno)
- Parkland: **200–250 MVU × 4 h**; progress **1–2 cm/h** on oxytocin